

CODUKU Project: Technology Stack Analysis

1. Architecture Overview

CODUKU is a sophisticated competitive coding platform designed for stability and high-performance code execution. It utilizes a microservices architecture combined with a production-grade execution engine to handle diverse programming languages securely.

2. Frontend Layer (User Experience)

- React (v18): Core library for building the dynamic user interface.
- Monaco Editor: The main coding engine (powering VS Code) with full syntax support.
- React Split / Lucide React: Providing the responsive multi-pane layout and modern visual assets.
- Vanilla CSS: A custom design system providing a premium 'Glassmorphism' feel.

3. Backend & Microservices

- FastAPI (Python): Powers asynchronous services (Judge, Auth, Leaderboard, Mentor).
- Flask: Legacy core handling main registration and question management.
- Judge Service: Acts as the orchestrator for all code evaluation logic.
- Mentor Service (Next.js/Chatbot): Standalone AI interface for student guidance.

4. Code Execution Engine

- Judge0: An open-source execution engine running 20+ languages in isolated containers.
- Docker: Handles container sandboxing to ensure server security during code execution.

5. Data Layer

- MongoDB: Flexible storage for profiles, test cases, and question banks.
- PostgreSQL: Relational storage for scoreboards and persistence for microservices.
- Redis: High-performance caching and task queuing for the judging system.
- ChromaDB: Vector storage used for AI search and documentation matching.

6. Infrastructure & Deployment

- NGINX: High-performance reverse proxy and API Gateway.
- Docker Compose: Service orchestration for managing the 10+ interconnected containers.
- Deployment Requirement: Recommended 8GB RAM Linux Host for full orchestration.